Understanding Developers' Experiences in Seeking Similar Rationale Information in Commit Messages

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ABSTRACT

Commit messages contain information about decisions taken and their underlying rationale, which can be valuable for solution reuse. However, effectively leveraging this information remains challenging as we lack a detailed understanding of developers' specific needs. Our work-in-progress addresses this challenge through a survey of developers' experiences seeking rationale information in commit messages. Here, we report preliminary analyses focused on needing and finding similar rationale information, suggesting that finding previous commits with similar rationale information (decisions and rationale) is important for completing development tasks, yet difficult and time-consuming in the current practice, which highlights the need for dedicated tool support. These insights can guide the design of future rationale-aware development tools.

CCS CONCEPTS

• **Software and its engineering → Software development techniques**.

KEYWORDS

Software rationale, Commit messages, Similar rationale information

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1 RESEARCH PROBLEM AND MOTIVATION

Rationale for changes in software projects is often recorded in commit messages [2]. Our prior work has proposed a vision for an end-to-end rationale reconstruction pipeline to empower rationale reuse [4], along with several automated analyses that enable such reuse—for example, by identifying similar rationale information in previous commits [3]. However, the usefulness of these analyses depends on whether developers actually need such information.

In this ongoing work, we propose to investigate developers' need for rationale information in depth. Specifically, we surveyed software developers on their practices and experiences with contextualizing rationale in commit messages.

In this paper, we present initial analyses and insights into **developers' experiences when needing and finding similar rationale information in earlier commit messages**. These insights will help us design tailored rationale analyses that better align with developers' needs, ultimately improving their productivity.

2 BACKGROUND AND RELATED WORK

Finding previous commits with similar rationale information can be useful for tasks like post-mortem analysis or example retrieval [1, 6, 7]. While prior studies explored developers search practices and experiences with rationale in commit messages [1], to the best of our knowledge, there is a lack of research on the broader dimensions of this specific need, including how frequently practitioners encounter it, the situations in which it arises, and the effort required to address it across different project roles (developers versus maintainers).

3 APPROACH

Survey Design and Pilot. Three researchers, two professors and a Ph.D. candidate, constructed the survey over three one-hour meetings, agreeing on 16 questions—6 on demographics—some inspired by prior work [1]. At the time, LLMs were not part of developer workflows. Thus, the questions –shown in Figure 1– excluded explicit mentions to AI usage. Four experienced graduate students and postdocs reviewed the survey, suggesting only minor edits.

Data Collection. The survey ran from April 4th to May 27th 2025. It was conducted online and anonymously, using Google Forms. We also created a basic landing page with study information, including the data policy. We used snowball sampling [5] to recruit participants. We advertised our survey on social media (LinkedIn, Mastadon and X), on community-specific mailing lists, on personal websites and during scientific events. Our survey targeted software developers and project maintainers specifically. Participation was voluntary with no compensation offered. We got 34 responses, all retained after manual validation.

4 RESULTS AND CONTRIBUTIONS

Demographics. Our participants represent a diverse population of open-source and closed-source developers and maintainers with different levels of experience. Open-source projects mentioned include maven, libssh2, nanofuzz, guix, linux kernel, kde and others. In the last question, respondents rated which task they spent more time on. Those focusing on writing commit messages were classified as developers, while others, including those spending equal time on both tasks, were classified as maintainers.

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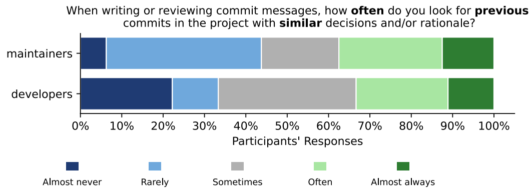
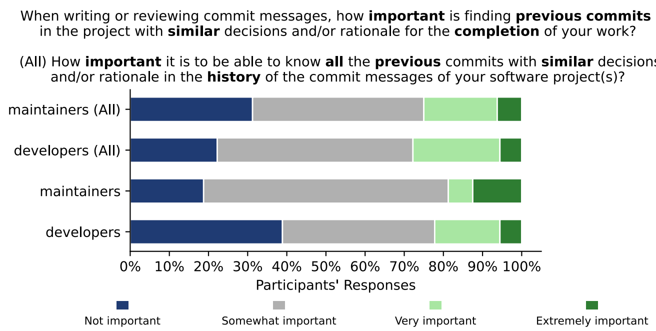
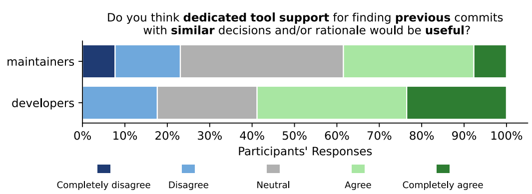
Experience of developers *needing* to find similar rationale information | Experience of developers *finding* similar rationale information



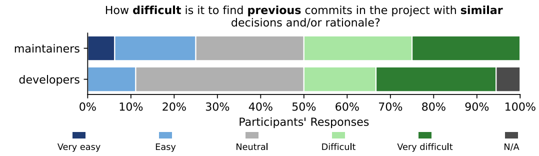
(a) Frequency of Need (Q7)



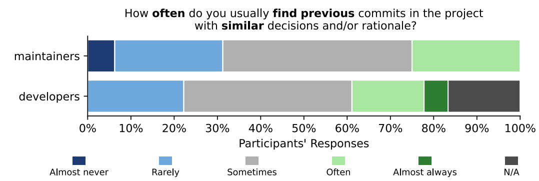
(b) Importance of Need (Q8, Q12)



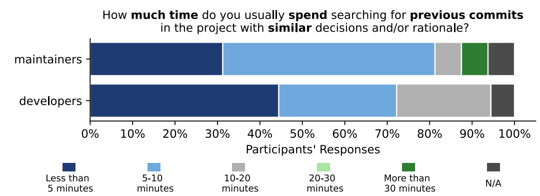
(c) Need for Dedicated Tool Support (Q16)



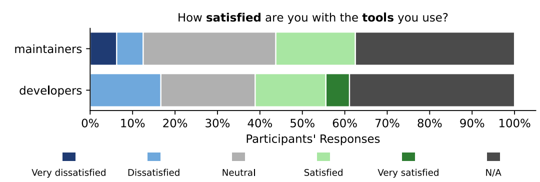
(d) Difficulty of Finding (Q9)



(e) Frequency of Finding (Q10)



(f) Search Time Until Finding (Q11)



(g) Experience with Current Tool Support (Q15)

Figure 1: Experience of developers *needing* to find, and *finding* similar rationale information in earlier commit messages

Results. To assess the *need* for finding similar rationale information, we examine three key aspects: how frequently developers encounter this need, how important they perceive it to be, and whether they express a need for dedicated tool support. First, participants reported needing to seek previous commits in the project with similar rationale information with diverse frequencies (Figure 1a). Overall, the majority (61,76%) need similar commits relatively frequently: sometimes, often or almost always, with maintainers reporting a more frequent need. Second, when asked about how important is it find previous commits with similar rationale information for the completion of your work, 70,58% of the participants reported relative importance: somewhat important, very important and extremely important. Similarly, 73,52% reported relative importance of finding all the previous commits with similar rationale information (Figure 1b). Maintainers expressed a stronger need to find previous commits containing similar rationale information, whereas developers emphasized the importance of retrieving all prior similar commits. Finally, most of the participants (52,94%) agreed (or completely agreed) that dedicated tool support would be useful (Figure 1c) with developers expressing a greater need.

To assess the experience of developers *finding* similar rationale information, we examine four key aspects: the perceived difficulty of the task, how frequently they succeed in finding such information, the time it takes to locate the needed information, and their experience with current tool support. First, almost half (48,48%) the responses indicate that finding similar rationale information can be difficult (or very difficult), while the 33.33% selected neutral difficulty (Figure 1d). Compared to developers, maintainers were more likely to find it easy to locate previous commits with similar rationale. Second, 45,16% of participants indicated partial success for finding similar commits (sometimes) with 29% succeeded rarely or almost never (Figure 1e). Maintainers reported less frequent success (Rarely or Almost never). Third, almost all participants (81,25%) spend less than 10 minutes on this task (Figure 1f). Maintainers generally spent less time than developers searching for similar commits. Finally, almost half the participants (52.9%) use tools to find similar rationale information, e.g., *git*, *gitlog* and *sourcetree* (Q13, Q14). 42,85% of our participants indicated neutral satisfaction while 23,8% expressed their dissatisfaction with the current tools (Figure 1g). Maintainers reported more dissatisfaction than developers.

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